

# Multi-Metric Text Similarity Evaluation for Contextual Sentence Reconstruction

# Motivation

- Resolve real time text concatenation to be a coherent stream of text
- Automatic transcription produces duplicate, redundant, or slightly misaligned sentence chunks
- Wanted it to be lightweight, deterministic
  - Not using LLM's to parse text
- Accurately detect false positive similar sentences

1. The quick brown fox jumped over the lazy dog
2. jumped over the lazy dog to get home quick
3. jumped over the lazy dog to get home quickly
4. so that he could go fishing.
5. Fishing, he could go soon, so that was it.

1. Concat first sentence
2. Overlapping exact text; Concat at indexes
3. Last word was corrected
  - This is the correct output to be used
4. Completely different; Concat
5. False positive when using similarity metrics

Final Output:

The quick brown fox jumped over the lazy dog to get home quickly so that he could go fishing. Fishing, he could go soon, so that was it.

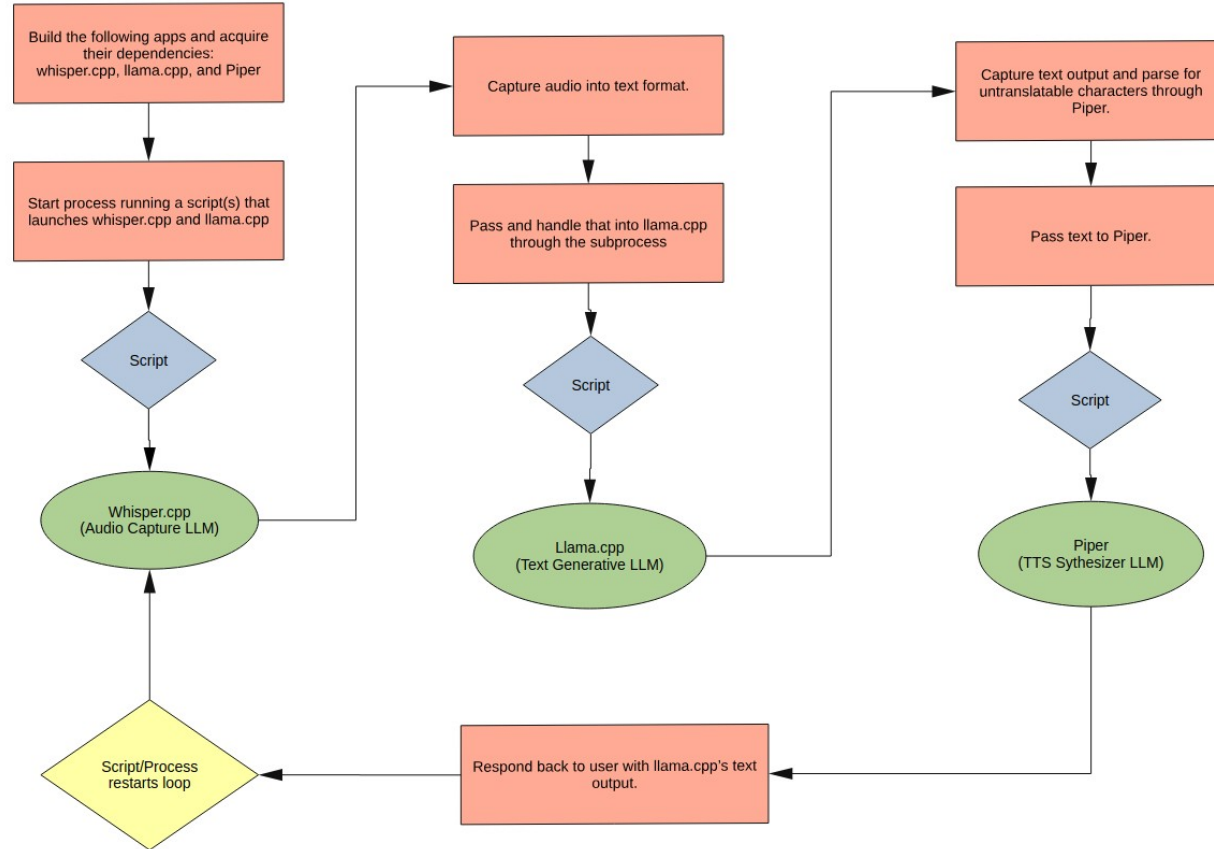
# Context for Problem

- whisper.cpp
  - Compiled terminal interface for OpenAI's Whisper automatic speech recognition (ASR) model
- llama.cpp
  - LLM interface of ggml tensors to run on local machines using GPU compute resources
- Piper
  - A TTS synthesiser LLM

# Purpose of these LLM's

- Initially used as a way to work with audio to text as a language speaking partner
- Discovered stream outputs do not consistently return accurate concatenated statements when running
  - Like an assembly line feed, you still need to manipulate the product
- Literature review confirmed real-time text concatenation is still an ongoing problem

# Flow Chart of Finding Problem



# Problem

- whisper.cpp output after processing is inconsistent
  - Duplicate text from stream
  - Redundancy from stream
  - Partial stream duplication
- The whole return stream was a lot of data that simply flowed out as the LLM processed audio in real time
  - One output could be sentence “YYZ”
  - Second could be “YYZABC”
  - Final output would be “YYZYYZABC”
  - We just wanted “YYZABC”
- Not as simple as a 1:1 string comparison

# Current Research Overview

- Algorithmic paradigms have become less the norm
- Greater reliance on LLM's to handle this task
  - High monetary cost, expensive to run, computational expensive as well
  - LLM's processing is excessive
- Many tests seem to only focus on one heuristic
  - Most research picks just one metric
  - Independent of other metrics like Jaccard, Rogue-I, etc.)
- Most research currently catered toward matching similarities
- Little research in:
  - Detecting false positives
  - Considering word order

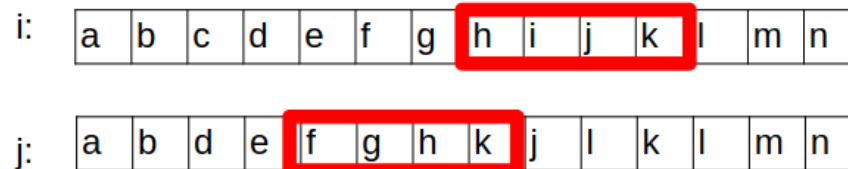
# Literature Review

- Multi-metric similarity for sentence merging
  - Uncommon and typically conducted independently
- Sliding window-based local context matching
  - Typically researches use a “global” approach
  - (Levenshtein, 1966; Smith & Waterman, 1981)
- False Positive Detection
  - Filtering out false positives is discussed but not always measured
  - (Vrbanec & Meštrović, 2023)
- Uses multi-tiered decision logic
  - Most research tests one heuristic



# Algorithm

- Use a fuzzy sliding window to compare local word chunks
  - Sliding window size of 4 for tests
- Similarity checks for Jaccard, Cosine, Dice
  - Rogue-I was not used in sliding window because that is for a total 1:1 comparison
- Compare all windows of both sentences until high similarity found
  - $O(n * m)$ 
    - Theoretically “poor”, but applicably fast
  - Finds a highly probable match between both sentences
- Relies on a set **Threshold** value to trigger a “found” similarity



```
# If high matches found, extremely similar.  
# If low matches found, implies it could be similar or there was an overlap.  
# If it's 0, it does not have an exact match anywhere within the windows.  
return window_matches_found / max(1, len(prev_list) - prev_win_size + 1)
```

# Algorithm Thresholds

- Percentage value that represents the base score
  - Called “Threshold Base” as the minimum value
  - Compounded by a multiplier for higher values
    - Did this because the spread would vary by the size of the threshold
- For certain conditions,  $>$  or  $<$  other values is desirable
  - Some values require higher or lower scores to surpass the threshold requirement

| Threshold | VERYLOW | LOW  | MEDIUM | HIGH | VERYHIGH |
|-----------|---------|------|--------|------|----------|
| 0.05      | 0.05    | 0.10 | 0.15   | 0.20 | 0.25     |
| 0.06      | 0.06    | 0.12 | 0.18   | 0.24 | 0.30     |
| 0.07      | 0.07    | 0.14 | 0.21   | 0.28 | 0.35     |
| 0.08      | 0.08    | 0.16 | 0.24   | 0.32 | 0.40     |
| 0.09      | 0.09    | 0.18 | 0.27   | 0.36 | 0.45     |
| 0.1       | 0.10    | 0.20 | 0.30   | 0.40 | 0.50     |
| 0.11      | 0.11    | 0.22 | 0.33   | 0.44 | 0.55     |
| 0.12      | 0.12    | 0.24 | 0.36   | 0.48 | 0.60     |
| 0.13      | 0.13    | 0.26 | 0.39   | 0.52 | 0.65     |
| 0.14      | 0.14    | 0.28 | 0.42   | 0.56 | 0.70     |
| 0.15      | 0.15    | 0.30 | 0.45   | 0.60 | 0.75     |
| 0.16      | 0.16    | 0.32 | 0.48   | 0.64 | 0.80     |
| 0.17      | 0.17    | 0.34 | 0.51   | 0.68 | 0.85     |
| 0.18      | 0.18    | 0.36 | 0.54   | 0.72 | 0.90     |
| 0.19      | 0.19    | 0.38 | 0.57   | 0.76 | 0.95     |
| 0.2       | 0.20    | 0.40 | 0.60   | 0.80 | 1.00     |
| 0.21      | 0.21    | 0.42 | 0.63   | 0.84 | 1.00     |
| 0.22      | 0.22    | 0.44 | 0.66   | 0.88 | 1.00     |
| 0.23      | 0.23    | 0.46 | 0.69   | 0.92 | 1.00     |
| 0.24      | 0.24    | 0.48 | 0.72   | 0.96 | 1.00     |
| 0.25      | 0.25    | 0.50 | 0.75   | 1.00 | 1.00     |
| 0.26      | 0.26    | 0.52 | 0.78   | 1.00 | 1.00     |
| 0.27      | 0.27    | 0.54 | 0.81   | 1.00 | 1.00     |
| 0.28      | 0.28    | 0.56 | 0.84   | 1.00 | 1.00     |
| 0.29      | 0.29    | 0.58 | 0.87   | 1.00 | 1.00     |
| 0.3       | 0.30    | 0.60 | 0.90   | 1.00 | 1.00     |
| 0.31      | 0.31    | 0.62 | 0.93   | 1.00 | 1.00     |
| 0.32      | 0.32    | 0.64 | 0.96   | 1.00 | 1.00     |
| 0.33      | 0.33    | 0.66 | 0.99   | 1.00 | 1.00     |
| 0.34      | 0.34    | 0.68 | 1.00   | 1.00 | 1.00     |
| 0.35      | 0.35    | 0.70 | 1.00   | 1.00 | 1.00     |

# Demonstration

[https://youtu.be/NBrxUxZb\\_FM](https://youtu.be/NBrxUxZb_FM)

# Evaluation of Tests

- Overall the tests aim to:
  - Test text similarity
  - Ensure accurate sentence concatenation and merging
  - Handle niche cases and catch false-positives
- Measure how well the system handles:
  - Near-duplicates
  - Exact duplicates
  - Reverse/reordered sentences
  - False positives that normal similarity metrics cannot infer
  - Natural speech variations and edge cases

# Test 1

- Contains general spoken text with partial or full duplicate lines
  - Slightly random
  - Like regular speech
- Tests realistic outputs from whisper.cpp processing

```
Hey how is it going today?  
How are you doing with your work?  
How are you doing with your very stressful work?  
How are you doing with your very stressful work that you have to do over  
this long weekend?  
How are things going at your job?  
I think talking into a microphone is very cool and that it is interesting  
because you can visually determine where audio is captured.  
What is captured is also another fascinating concept.  
I think talking into recording devices is cool and that it is interesting.  
Hey how is it going today? I think  
think talking into a microphone is very cool and that it is interesting  
yeah  
very cool and that it is interesting because you can visually  
because you can visually determine where ah...  
because you can visually determine where audio  
is captured. What is captured is also another fascinating concept.  
I think talking into recording devices is cool and that it is interesting.  
Hey how is it going today?  
Hey how is it going today?  
it going today?  
today? I think that this is very interesting don't you?  
And this is also cool.
```

# Test 2-4

- Test 2
  - Only last few words were different
- Test 3
  - Contains exact duplicates
- Test 4
  - All lines are unique
- Ignores exact duplicates and retains only unique lines
- Appends different lines entirely

```
The quick brown fox jumps over the lazy dog near the river.  
The quick brown fox jumps over the lazy dog in the sunny meadow.  
The quick brown fox jumps over the lazy dog without hesitation.  
The quick brown fox jumps over the lazy dog to catch its prey.  
The quick brown fox jumps over the lazy dog before sunset.
```

```
The weather today is sunny and warm.  
The weather today is sunny and warm.  
I went to the park this afternoon.  
I went to the park this afternoon.  
Walking through the trees was very relaxing.  
Walking through the trees was very relaxing.  
The birds were singing beautifully.  
The birds were singing beautifully.
```

```
The sun was shining bright.  
I saw children playing happily.  
The gentle breeze cooled the air.  
I sat on a bench and read my book.  
The flowers were blooming in vibrant colors.  
The birds were singing beautifully.  
I went to the park this afternoon.  
The weather today is sunny and warm.  
Walking through the trees was very relaxing.
```

# Test 5

- Each sentence heavily overlaps with the previous one
- Intended to trigger overwrites at specific text indexes
- Tests accurate partial appends and efficient merging

I was thinking about how memory affects decision making.  
about how memory affects decision making and emotional responses.  
decision making and emotional responses can be deeply intertwined.  
can be deeply intertwined with personal experiences from the past.  
with personal experiences from the past, like childhood or trauma.  
past, like childhood or trauma, can really impact how we respond.  
impact how we respond to stress and unexpected challenges.  
how we respond to stress and unexpected challenges depends on resilience.  
stress and unexpected challenges depends on resilience and coping strategies.  
depends on resilience and coping strategies we build over time.  
coping strategies we build over time shape our behavior.  
we build over time shape our behavior and reactions to change.  
shape our behavior and reactions to change in dynamic environments.  
reactions to change in dynamic environments test our adaptability.  
dynamic environments test our adaptability and mindset under pressure.  
test our adaptability and mindset under pressure and uncertainty.  
mindset under pressure and uncertainty reveals deeper personality traits.  
under pressure and uncertainty reveals deeper personality traits and beliefs.  
personality traits and beliefs guide how we interpret experiences.  
how we interpret experiences defines the stories we tell ourselves.

# Test 6-8

- Variations of Test 5:
  - Sentence forward, next is the same sentence backward
  - All sentences in reverse
  - Flipping sentence order from last to first
- Tests to trigger false positives
- Somewhat realistic at times as English speakers can state something grammatically incorrect
  - Examples:
    - Yoda speak
    - Poetry

I was thinking about how memory affects decision making.  
making decision affects memory how about thinking was I.  
decision making and emotional responses can be deeply intertwined.  
intertwined deeply be can responses emotional and making decision.  
can be deeply intertwined with personal experiences from the past.

making decision affects memory how about thinking was I.  
responses emotional and making decision affects memory how about.  
intertwined be deeply can responses emotional and making decision.  
past the from experiences personal with intertwined deeply be can.  
trauma or childhood like past the from experiences personal with.

with personal experiences from the past, like childhood or trauma.  
can be deeply intertwined with personal experiences from the past.  
decision making and emotional responses can be deeply intertwined.  
about how memory affects decision making and emotional responses.  
I was thinking about how memory affects decision making.

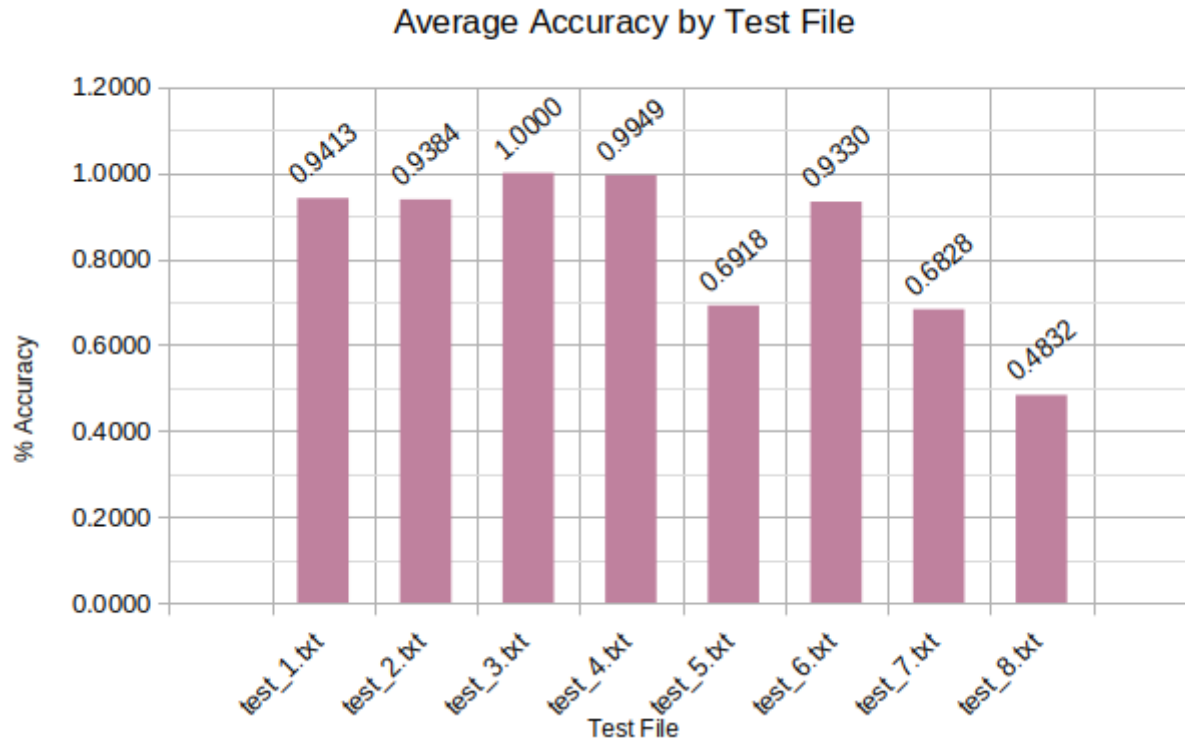


# Test Scores Explained

- The final appended result (as one full string) compared against an answer file
  - Final algorithm output compared to answer text
- Leveraged Rogue-I scoring
  - Finds the longest sequence of words in order between two lists
    - Takes proportional list lengths and sequential words found into consideration
  - Like the game “Mastermind”, but proportional to length

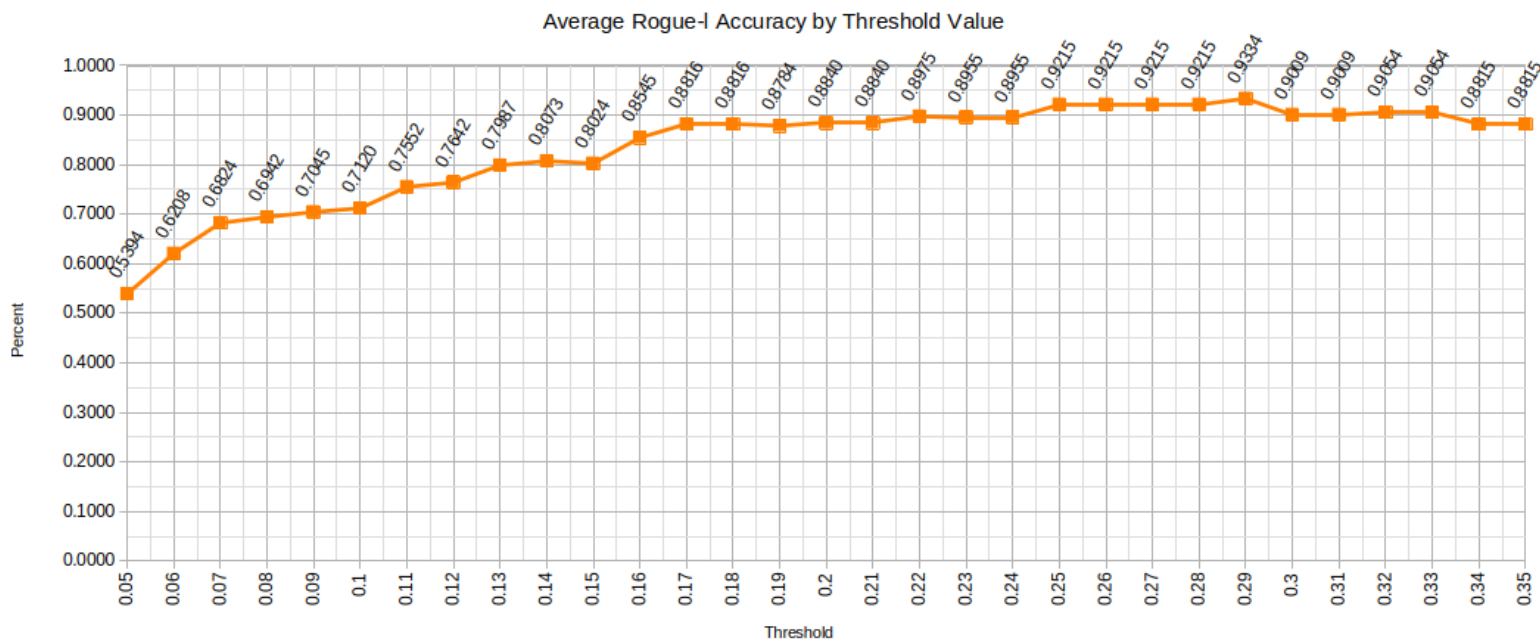


| Threshold | Test File  | Answer File  | Final Result |
|-----------|------------|--------------|--------------|
| 0.05      | test_1.txt | answer_1.txt | 0.6636       |
| 0.05      | test_2.txt | answer_2.txt | 1.0000       |
| 0.05      | test_3.txt | answer_3.txt | 1.0000       |
| 0.05      | test_4.txt | answer_4.txt | 0.9212       |
| 0.05      | test_5.txt | answer_5.txt | 0.2151       |
| 0.05      | test_6.txt | answer_6.txt | 0.3077       |
| 0.05      | test_7.txt | answer_7.txt | 0.1183       |
| 0.05      | test_8.txt | answer_8.txt | 0.0891       |
| 0.06      | test_1.txt | answer_1.txt | 0.7336       |
| 0.06      | test_2.txt | answer_2.txt | 1.0000       |
| 0.06      | test_3.txt | answer_3.txt | 1.0000       |
| 0.06      | test_4.txt | answer_4.txt | 0.9212       |
| 0.06      | test_5.txt | answer_5.txt | 0.2151       |
| 0.06      | test_6.txt | answer_6.txt | 0.8889       |
| 0.06      | test_7.txt | answer_7.txt | 0.1183       |
| 0.06      | test_8.txt | answer_8.txt | 0.0891       |
| 0.07      | test_1.txt | answer_1.txt | 0.9680       |
| 0.07      | test_2.txt | answer_2.txt | 1.0000       |
| 0.07      | test_3.txt | answer_3.txt | 1.0000       |
| 0.07      | test_4.txt | answer_4.txt | 1.0000       |
| 0.07      | test_5.txt | answer_5.txt | 0.3400       |
| 0.07      | test_6.txt | answer_6.txt | 0.8588       |
| 0.07      | test_7.txt | answer_7.txt | 0.2034       |
| 0.07      | test_8.txt | answer_8.txt | 0.0891       |
| 0.08      | test_1.txt | answer_1.txt | 0.9680       |
| 0.08      | test_2.txt | answer_2.txt | 1.0000       |
| 0.08      | test_3.txt | answer_3.txt | 1.0000       |
| 0.08      | test_4.txt | answer_4.txt | 1.0000       |
| 0.08      | test_5.txt | answer_5.txt | 0.4340       |
| 0.08      | test_6.txt | answer_6.txt | 0.8588       |
| 0.08      | test_7.txt | answer_7.txt | 0.2034       |
| 0.08      | test_8.txt | answer_8.txt | 0.0891       |



| Threshold | Test File  | Answer File  | Average Accuracy by Test File |
|-----------|------------|--------------|-------------------------------|
| 0.05      | test_1.txt | answer_1.txt | 0.9413                        |
| 0.05      | test_2.txt | answer_2.txt | 0.9384                        |
| 0.05      | test_3.txt | answer_3.txt | 1.0000                        |
| 0.05      | test_4.txt | answer_4.txt | 0.9949                        |
| 0.05      | test_5.txt | answer_5.txt | 0.6918                        |
| 0.05      | test_6.txt | answer_6.txt | 0.9330                        |
| 0.05      | test_7.txt | answer_7.txt | 0.6828                        |
| 0.05      | test_8.txt | answer_8.txt | 0.4832                        |

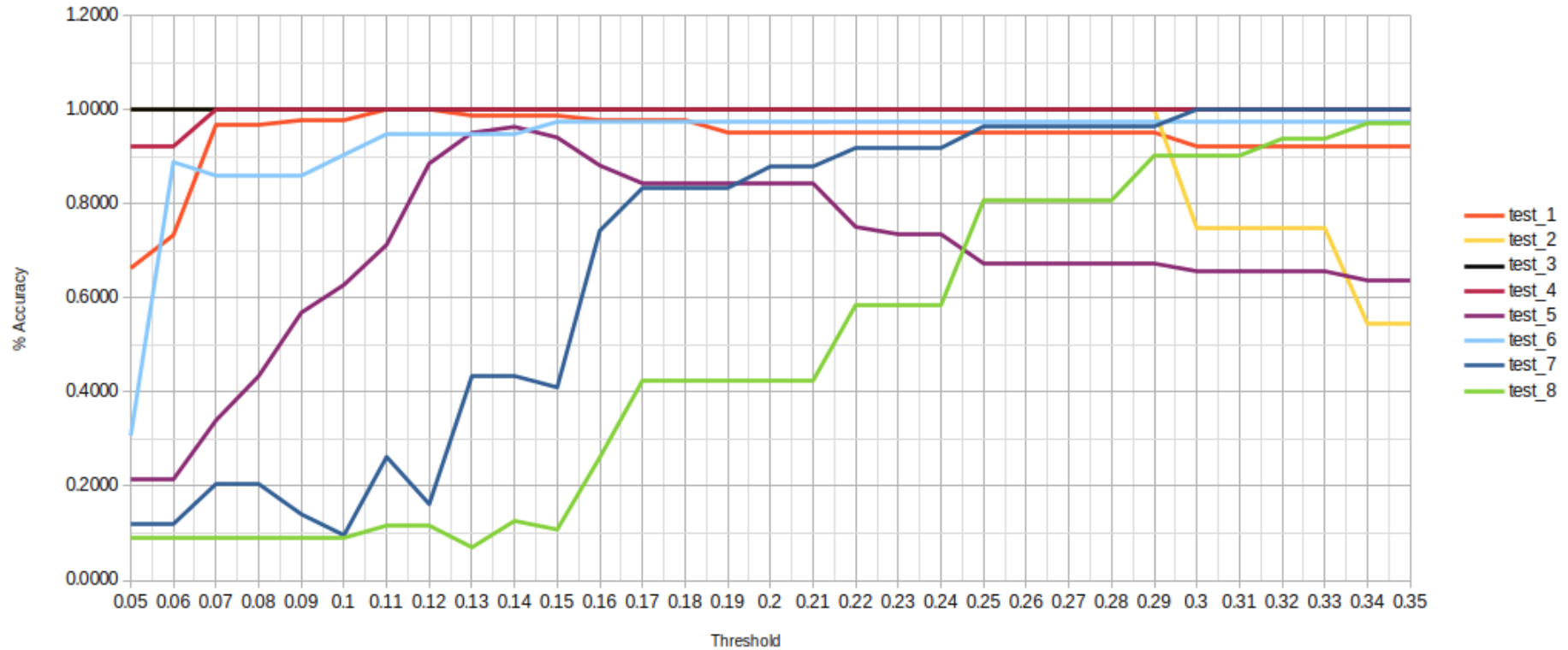
- Sum of accuracy of each test by threshold, divided by number of threshold tests (31)



| Threshold | Test File  | Answer File  | Average Rouge-I Accuracy<br>by average of results<br>by threshold |
|-----------|------------|--------------|-------------------------------------------------------------------|
| 0.05      | test_1.txt | answer_1.txt | 0.5394                                                            |
| 0.06      | test_1.txt | answer_1.txt | 0.6208                                                            |
| 0.07      | test_1.txt | answer_1.txt | 0.6824                                                            |
| 0.08      | test_1.txt | answer_1.txt | 0.6942                                                            |
| 0.09      | test_1.txt | answer_1.txt | 0.7045                                                            |
| 0.1       | test_1.txt | answer_1.txt | 0.7120                                                            |
| 0.11      | test_1.txt | answer_1.txt | 0.7552                                                            |
| 0.12      | test_1.txt | answer_1.txt | 0.7642                                                            |
| 0.13      | test_1.txt | answer_1.txt | 0.7987                                                            |
| 0.14      | test_1.txt | answer_1.txt | 0.8073                                                            |
| 0.15      | test_1.txt | answer_1.txt | 0.8024                                                            |
| 0.16      | test_1.txt | answer_1.txt | 0.8545                                                            |
| 0.17      | test_1.txt | answer_1.txt | 0.8816                                                            |
| 0.18      | test_1.txt | answer_1.txt | 0.8816                                                            |
| 0.19      | test_1.txt | answer_1.txt | 0.8784                                                            |
| 0.2       | test_1.txt | answer_1.txt | 0.8840                                                            |
| 0.21      | test_1.txt | answer_1.txt | 0.8840                                                            |
| 0.22      | test_1.txt | answer_1.txt | 0.8975                                                            |
| 0.23      | test_1.txt | answer_1.txt | 0.8955                                                            |
| 0.24      | test_1.txt | answer_1.txt | 0.8955                                                            |
| 0.25      | test_1.txt | answer_1.txt | 0.9215                                                            |
| 0.26      | test_1.txt | answer_1.txt | 0.9215                                                            |
| 0.27      | test_1.txt | answer_1.txt | 0.9215                                                            |
| 0.28      | test_1.txt | answer_1.txt | 0.9215                                                            |
| 0.29      | test_1.txt | answer_1.txt | 0.9334                                                            |
| 0.3       | test_1.txt | answer_1.txt | 0.9009                                                            |
| 0.31      | test_1.txt | answer_1.txt | 0.9009                                                            |
| 0.32      | test_1.txt | answer_1.txt | 0.9054                                                            |
| 0.33      | test_1.txt | answer_1.txt | 0.9054                                                            |
| 0.34      | test_1.txt | answer_1.txt | 0.8815                                                            |
| 0.35      | test_1.txt | answer_1.txt | 0.8815                                                            |

- Sum of accuracy of all tests with a specific threshold, divided by number of tests (8)

Accuracy of Test by Threshold using Rogue-I

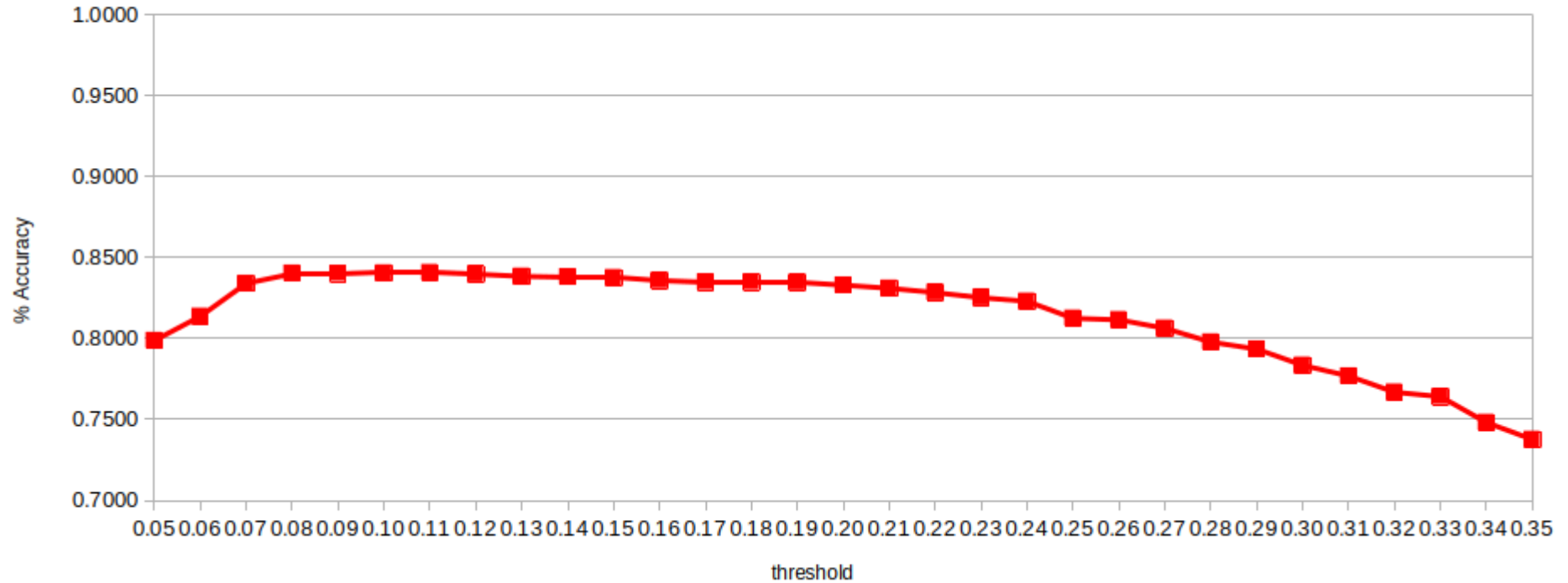


- Rogue-I score across all tests and thresholds
  - Tests show higher base thresholds average a higher level of accuracy

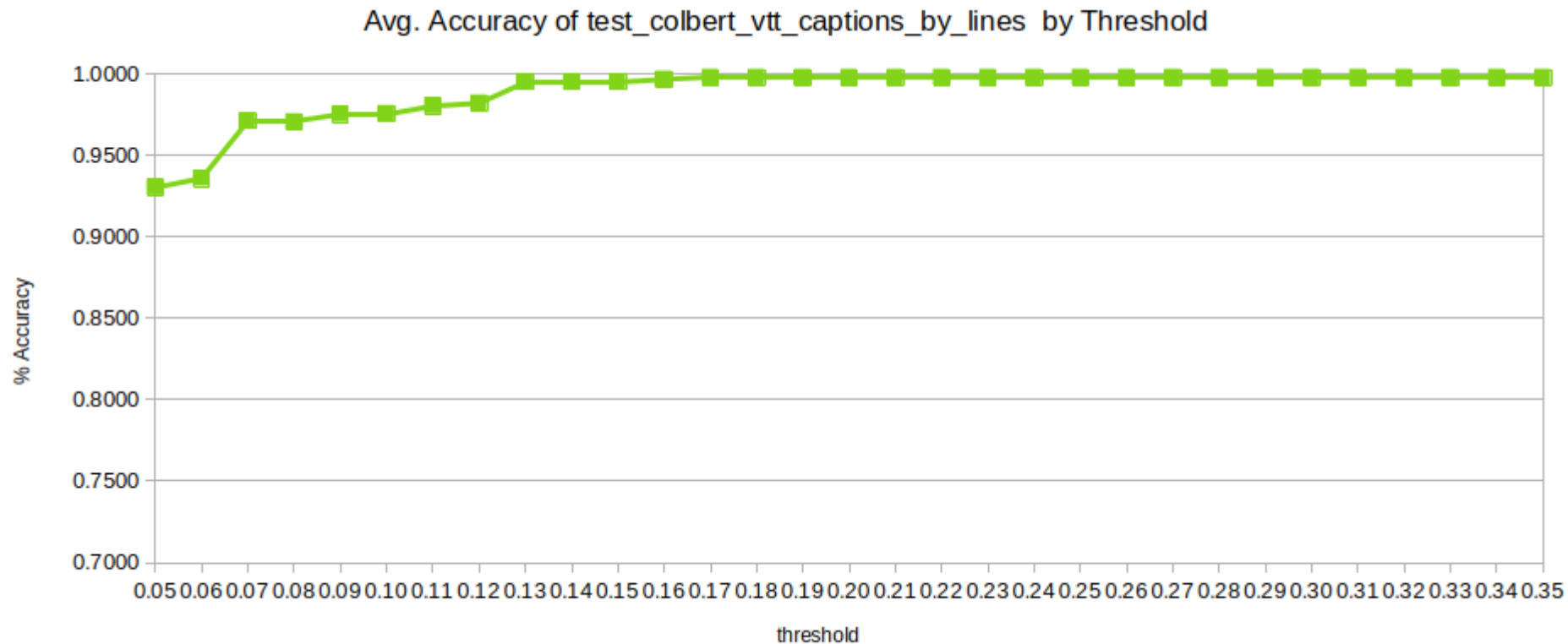
# Colbert Tests

- Tested against the Stephen Colbert's White House Correspondants Dinner Speech
  - ~24 minutes long
  - Stress testing of my algorithm
- Compared against the Youtube .vtt subtitles file to use as a control for comparing against the actual closed captions
- Humor and out of context jokes, political concepts, diplomatic “higher register” speaking, casual slang, etc.

Avg. Accuracy of test\_colbert\_whisper\_captured\_lines.txt by Threshold



- Shows accuracy against Colbert Whisper text output
  - Varied text length, partial or full repeating outputs captured
  - Real world scenario



- Shows accuracy against Colbert VTT (Youtube Subtitle File)
  - Accuracy reflects results obtained from tests
  - Line lengths were consistent, no repeating unless audio actually did repeat

# Results

- Lower threshold values resulted in higher accuracy scores for “real world”-like tests
  - Test 1 and Colbert Test
- Higher threshold values resulted in higher accuracy scores for tests that would result in several false positives
  - Tests 5-8
- Still not perfect due to enumerable permutations of NLP



# Results (cont'd.)

- Higher scores in Colbert tests caused higher threshold necessary to succeed leading to more evaluations to be interpreted as false positives
  - Why lower scores were more successful in the real world test
- Higher scores in Test files did not have this problem
  - Likely due to the less random, shorter, simpler types of test
  - Likely not enough length and variability in text

# Conclusion

- Effective Alternative to LLMs
  - Combines multiple lexical and structural similarity metrics
    - Jaccard, Cosine, Dice, Rogue-I
  - Lightweight sentence merging
  - Can be adjusted and fine tuned
- Dynamic Thresholds
  - Tiered threshold logic accurately filters false positives
  - Supports append, overwrite at specified indexes, ignore
- Sliding window comparison captures local context
- Fuzzy matching tolerates imperfections

# Future Outlook

- Provides an alternative to LLMs for captioning and real time string appending handling
- Great for live closed-captioning systems
  - Use algorithm for concatenating live realtime audio
  - Current CC systems still stream duplicate info per line
- Captioned Telephone service
  - Still requires humans for transcription
  - My algorithm could be used to reduce manual human transcription

# Special Thanks

- My Advisor - Dr. Nguyen
- UW-Whitewater instructors and faculty

